

1 Propagations of Uncertainty

Relates the uncertainty of a calculated result, r , to the uncertainties in the input variables

Two approaches will be used:

1. Kline-McClintock(KM) -Analytical approaches
2. Monte Carlo (MC) -Purely computational approach (Preferred)

1.1 Kline-McClintock

Suppose:

$$r = f(x_1, x_2, x_3, x_L)$$

Then an uncertainty interval for r is:

$$r = \bar{r} \pm k_r u_r (P\text{percent sign})$$

$$k_r \equiv \text{coverage factor} = \lim_{v \rightarrow \infty} t_{v,p}$$

$$\bar{r} = f(\bar{x}_1, \bar{x}_2, \dots, \bar{x}_L)$$

$$u_r = \sum_{i=1}^L ((\phi_{x_i} u_{x_i})^2)^{1/2} = \text{std uncertainty in } R$$

$$\phi_{x_i} = \left. \frac{df}{dx_i} \right|_{x_i = \bar{x}_i} = \text{sensitivity index}$$

1.2

For the project, need to calculate $\dot{m}$ UI's for three cases
using NEED REFERENCE HERE

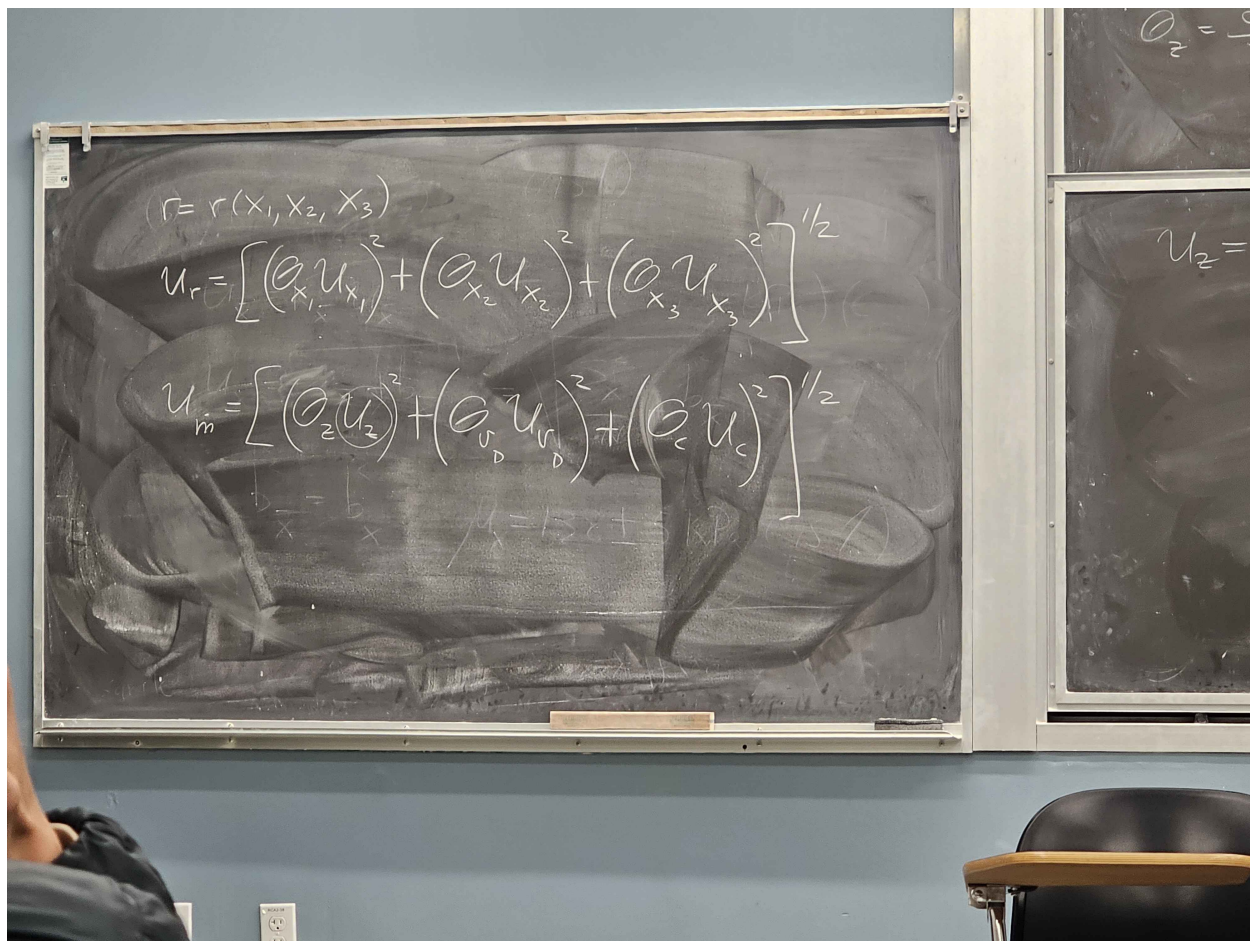
1.3 Example

$$r = r(x_1, x_2, x_3)$$

$$u_{\dot{m}} = [(\phi_z u_z)^2 + (\phi_{v_d} u_{v_d})^2 + (\phi_c u_c)^2]^{1/2}$$

$$\phi_{x_1} = \frac{df}{dx_i} = \frac{df}{dz} = \frac{d}{dz} [cz^{1/2} v_d^{-1/2}] = \phi_z$$

$$\phi_z = \left. \frac{1}{2} cz^{-1/2} v_d^{-1/2} \right|_{z=\bar{z}, v_d=\bar{v}_d, c=\bar{c}} = \frac{1}{2\bar{c}} \frac{1}{\bar{z}^{1/2} \bar{v}_d^{1/2}}$$



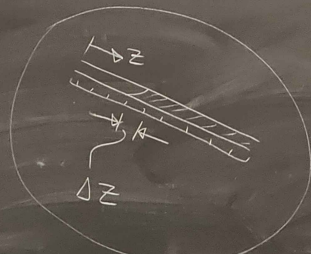
$$\left[\left(\frac{\partial u}{\partial x_2} \right)^2 + \left(\frac{\partial u}{\partial x_3} \right)^2 \right]^{1/2}$$

$$\left[\left(\frac{\partial u}{\partial x_D} \right)^2 + \left(\frac{\partial u}{\partial x_C} \right)^2 \right]^{1/2}$$

$$\theta_{x_1} = \frac{\partial f}{\partial x_1} = \frac{\partial f}{\partial \bar{z}} = \frac{\partial}{\partial \bar{z}} \left[C \bar{z}^{1/2} \bar{v}_D \right]$$

$$\theta_z = \frac{1}{2} C \bar{z}^{-1/2} \bar{v}_D^{-1/2} \Big| = \frac{1}{2} C \frac{1}{\bar{z}}$$

$$\theta_z = \frac{\partial \bar{m}}{\partial \bar{z}} \quad \begin{matrix} \bar{z} = \bar{z} \\ \bar{v}_D = \bar{v}_D \\ C = C \end{matrix} \quad \theta_z = \frac{1}{2}$$

$$u_z = (s_z^2 + b_z^2)^{1/2}$$


$$\frac{\partial f}{\partial z} = \frac{\partial}{\partial z} \left[C \bar{z}^{1/2} \bar{v}_D^{-1/2} \right] = \phi_z$$

$$\left. \bar{v}_D^{-1/2} \right| = \frac{1}{\bar{z}} C \frac{1}{\bar{z}^{1/2} \bar{v}_D^{1/2}}$$

$$\begin{aligned} \bar{z} &= \bar{z} \\ \bar{v}_D &= \bar{v}_D \\ C &= C \end{aligned}$$

$$\phi_z = \frac{1}{\bar{z}} (0.0504)$$

$$\frac{1}{8^{1/2} (0.858)^{1/2}} = 0.0096 \frac{\text{kg}}{\text{mm H}_2\text{O}}$$

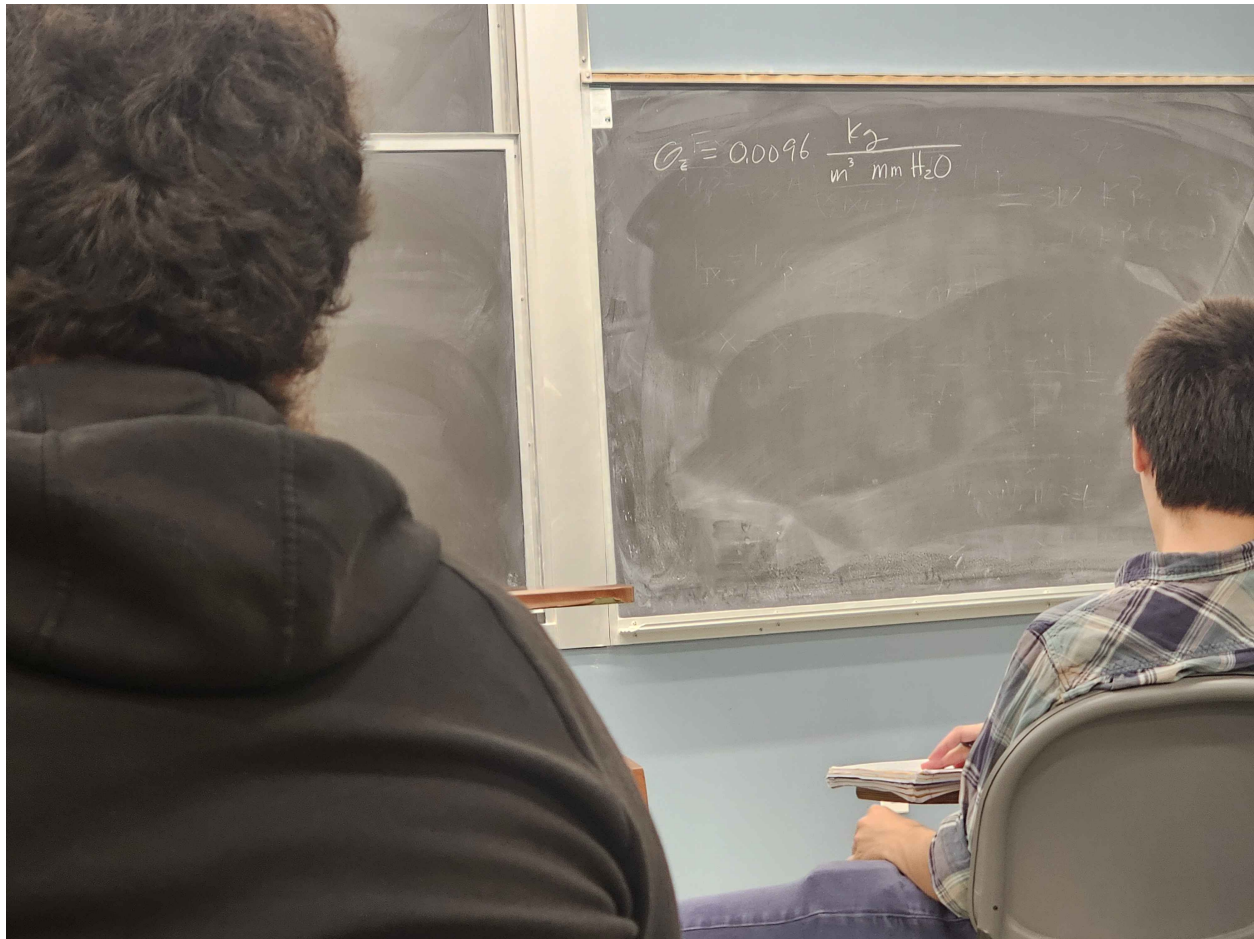
$$\bar{C} = 0.0504$$

$$\bar{z} = 8 \text{ mm H}_2\text{O}$$

$$\bar{v}_D = 0.858 \frac{\text{m}^3}{\text{kg}}$$

$\bar{v}_D$ = Value from the psychrometric chart (or calculated)

$$\frac{\text{kg}}{\text{mm H}_2\text{O}}$$



2 Psychrometric chart reading error

3 Obstruction meter theory

$$Q_1 = K_0 A_0 \left(\frac{2\Delta P}{\rho} \right)^{1/2} Y$$

$$\dot{m} = \rho_1 Q_1 = K_0 A_0 (2\Delta P \rho_1)^{1/2} Y$$

$$\dot{m} = f(K_0, A_0, \rho_1, \Delta P, Y)$$